Lars Larsen

Electronics Design Engineer — Analog, Mixed-Signal & Video Hardware

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SUMMARY

Electronics engineer with 15 years designing analog, mixed-signal and video hardware that ships. Lead designer of LZX Industries' 100+ products: discrete video-rate analog (Gilbert-cell and matched-pair multipliers, keyers, sync), video I/O from composite to HDMI, and the FPGA and MCU systems that control them. Schematic, simulation, layout direction, DFM, bring-up and test.

EXPERIENCE

Founder & Principal Electronics Engineer — LZX Industries

Portland, OR · 2011 – Present

Video synthesizer and video-processing instrument maker (DBA 2011, LLC 2017). Lead designer of every product in a three-person core team (partners on firmware/HDL and board layout).

- Led 100+ hardware products from concept to production across six Eurorack series, standalone instruments, DIY kits, cases and power: ~240 production circuit-board designs and ~140 metal front panels; $6.4M direct-to-consumer sales (2018–2026) plus a dealer channel.
- Videomancer (shipping since late 2025): architected a standalone FPGA video synthesizer — iCE40 HX4K + RP2040; ADV7611/ADV7513 HDMI in/out; ADV7181C + ADA4412-3 analog input; ADV7393 SD/ED/HD encoder; composite, S-Video, YPbPr and RGB — with user-loadable effects in Ed25519-signed packages, 69 public release candidates, and an open GPL-3.0 SDK.
- Chromagnon (RevJ prototype on the bench): led a video-rate analog/digital instrument through seven board revisions to an 8-layer core; replaced the Zynq SoM with iCE40 + RP2040 driving an analog Gilbert-cell XY engine, after costing an all-digital dual-ECP5 alternative — on-board silicon cut 64% ($195 to $70), BOM a further 26%.
- Analog video design for shipped product lines: matched-pair multipliers and blenders, keyers, colorspace mapping, sync separation and video-rate op-amp stages; co-defined the LZX 1V signal convention (output stages, 100 kΩ inputs) used across every Eurorack series.
- Gen3 power-entry platform (shipped, 2021–2023 revisions): architected FPGA12, SYNC12, PSU8 and PSU16 — a single +12 V input, diode-OR'd and EMI-filtered, converted to ±5.5 V by a synchronous buck and an inverting converter, then post-regulated to clean ±5 V by RF LDOs for analog video, with enable-chained iCE40 rails and ferrite-isolated clock/sync supplies; used across ~15 Eurorack modules and TBC2.
- Discrete high-speed analog: Chromagnon's XY engine (an affine transform of video-rate scan ramps) uses MC1496 Gilbert-cell rotators, discrete matched-pair two-quadrant multipliers and VCAs, diode-sigmoid keying and ±5 V 130 MHz op-amp stages; rightsized from 28 multiplier cells with 56 isolated supply rails to 8 (–190 placements).
- Simulation (2026, AI-assisted tooling): ngspice reviews with derating rules and IBIS-derived driver models — a ~200-check pre-fab review of a clock board found 9 failures fixed before fabrication; ~60-part shared model library.
- Mechanical: sheet-metal enclosures (Videomancer, built by HLH Prototypes; Eurorack cases and bus shields via SendCutSend), injection-molded custom knobs, and ~140 metal front panels, modeled in Inventor and FreeCAD.
- Manufacturing: contract-manufactured the line with ~40 active suppliers (JLCPCB, HLH Prototypes, SendCutSend, DigiKey); 1,281 manufacturing orders in MRPeasy, migrated to InvenTree/ERPNext in 2026; directed board layout by a partner.

Operations Manager — Darkplace Manufacturing

Portland, OR · 2015 – 2017

Manufacturing arm for the LZX Industries product line.

- Managed operations, purchasing and customer communication for a manufacturing team of 10+ people building the LZX product line, while continuing as its product designer.

Earlier

- Multimedia Programmer, University of North Texas, Denton TX — 2008–2010, 2011–2015

- Lead Service Technician, Switched On (synthesizer sales & repair), Austin TX — 2010
- New Media Developer, FUNimation Entertainment, Flower Mound TX — 2007–2008
- Multimedia Developer, TargetScope, Plano TX — 2006–2007

SELECTED PROJECTS

- **Chromagnon web simulator** — In-browser WebGL port of the Chromagnon instrument model, checked against golden vectors. lzxindustries.net/instruments/chromagnon/tools/simulator
- **LZX documentation** — Manuals, patch guides and specifications for the product line (open source, github.com/lzxindustries/lzxdocs). docs.lzxindustries.net
- **Cadet open hardware** — DIY video-synth module kits, open-sourced in 2022 with schematics, layouts and BOMs. github.com/lzxindustries/lzxcadet

SKILLS

Electronics	Analog/mixed-signal and video (SD–1080p, composite/S-Video/YPbPr/RGB/HDMI, genlock), discrete high-speed analog, power distribution, ngspice/SPICE and IBIS, DipTrace, Eagle, KiCad + own fork, DFM, BOM cost engineering, bring-up, oscilloscope/SCPI automation
Firmware & FPGA	C/C++ (C++17/20, freestanding, no-heap), Zynq-7000 bare-metal and FreeRTOS, STM32F4, RP2040/RP2350, AVR, TinyUSB, Ed25519 signing; VHDL, Verilog, GHDL, VUnit, Yosys/nextpnr (iCE40; ECP5 designs), Vivado (Zynq-7000)
Mechanical	Inventor, FreeCAD, build123d, sheet-metal enclosure design, injection-molded knobs, front-panel graphics, Blender product rendering
Product & operations	Concept-to-production ownership, PLM (MRPeasy, InvenTree), ERP, supplier and contract-manufacturer management, pricing, documentation, open-source hardware programs
Software	Python (NumPy, OpenCV, PySide6), TypeScript/React (Shopify Hydrogen/Remix, Vitest, Playwright), Docker, Forgejo CI, Linux servers, AI-assisted development (Claude Code, Cursor, MCP)

EDUCATION

- B.A., Radio, Television & Film — University of North Texas, 2006